

Applications of Database Systems

Supermarket Purchases

When you buy items at a supermarket, the checkout process uses a **database**.

1. The checkout assistant scans your items using a barcode reader.
2. The barcode is linked to an application program that retrieves the item's price from a **product database**.
3. The program updates the inventory, reducing the number of items in stock, and displays the price.
4. The system may automatically reorder items if the stock falls below a certain level.
5. If a customer calls the supermarket, an assistant can check item availability using an application program connected to the database.

Credit Card Purchases

When you use a credit card, the system checks your credit limit via a database.

1. The assistant checks your credit using a telephone or card reader linked to a computer system.
2. A **database** contains your purchase history.
3. An application program uses your credit card number to verify the purchase amount.
4. After confirmation, the purchase details are added to the **database**.
5. The program checks if the card is on a stolen or lost list.
6. Other applications generate monthly statements and credit accounts upon payment.

Holiday Booking at Travel Agents

When booking a holiday, travel agents access multiple databases for holiday and flight details.

1. The agent checks databases for holiday and flight information.
2. The system makes necessary booking arrangements.
3. The system prevents overbooking, ensuring that two agents don't book the same holiday or flight seat.
 - If only one seat remains on a flight and two agents try to book it, the system allows one booking and informs the other agent of the unavailability.
4. The travel agent may use a separate database for invoicing.

Using the Local Library

Libraries use databases to manage books, readers, and reservations.

1. The library has a **database** with details of books, readers, and reservations.
2. A computerized index allows readers to find books by title, author, or subject.
3. The system manages reservations, notifying readers when books are available.
4. The system sends reminders for overdue books.

Traditional File-Based Systems

Definition of Data

Data is a collection of related facts and figures that can be processed to produce meaningful information.

For example, a property for rent file contains records for each property. Each record has fields representing characteristics like address, property type, and number of rooms.

Limitations of File-Based Systems

The following table summarizes the limitations of file-based systems:

Limitation	Description
Separation and Isolation of Data	Data is difficult to access when stored in separate files.
Duplication of Data	Uncontrolled duplication leads to wasted storage and potential inconsistencies.
Data Dependence	The physical structure and storage of data files are defined in the application code.
Incompatible File Formats	N/A
Fixed Queries	N/A
Proliferation of Applications	N/A

Separation and Isolation of Data

Accessing data across separate files is difficult.

- Example: Listing houses that match client requirements requires creating a temporary file of clients and searching the property file, which is complex.
- Application developers must synchronize file processing, especially when data is needed from multiple files.

Duplication of Data

File-based systems encourage uncontrolled data duplication due to their decentralized nature.

- Example: Property and client details are duplicated in sales and contracts departments.
- Duplication is wasteful, costing time, money, and storage space.
- Data integrity is compromised, leading to inconsistencies.

Data Integrity Issues

Duplication can cause inconsistencies.

- Example 1: If an employee's address change is only updated in personnel but not payroll, pay slips will be sent to the wrong address.
- Example 2: If an employee's promotion and salary increase are updated in personnel but not payroll, the employee will receive the wrong salary.
- These inconsistencies require time and effort to resolve.
- Without automatic updates, errors are likely, even if changes are communicated.

Data Dependence

The structure and storage of data files are defined in the application code.

Relational DBMS Products in the 1980s

Examples include:

- DB2
- SQL/DS from IBM
- Oracle from Oracle Corporation.

There are now several hundred relational DBMS products.